

Lecture 2, September 25th, 2025

1. Bayesian inference: general formulation + examples
2. Monte Carlo algorithms

Bayesian inference (generic setup)

Inputs

- $\underline{y}$ — vector of data (could be a matrix or a tensor)
- $\underline{\theta}$ — vector of model parameters (e.g., θ — proportion)
- $p(\underline{y} | \underline{\theta})$ — density/prob defined by the data generating model (e.g., $\Pr(y|\theta)$ — Binomial)
- $p(\underline{\theta})$ — density of the prior distribution that encodes our prior beliefs about $\underline{\theta}$ (e.g., $p(\theta)$ — Beta density)

Output: $p(\underline{\theta} | \underline{y}) = \frac{p(\underline{y} | \underline{\theta}) p(\underline{\theta})}{\int p(\underline{y} | \underline{\theta}) p(\underline{\theta}) d\underline{\theta}}$ — posterior density of model parameters

In practice, we use the following summaries of the posterior distribution:

- 1) point estimation: median, mode, or mean of $p(\underline{\theta} | \underline{y})$
- For example,

$$\hat{\underline{\theta}} = E(\underline{\theta} | \underline{y}) = \int \underline{\theta} p(\underline{\theta} | \underline{y}) d\underline{\theta}$$

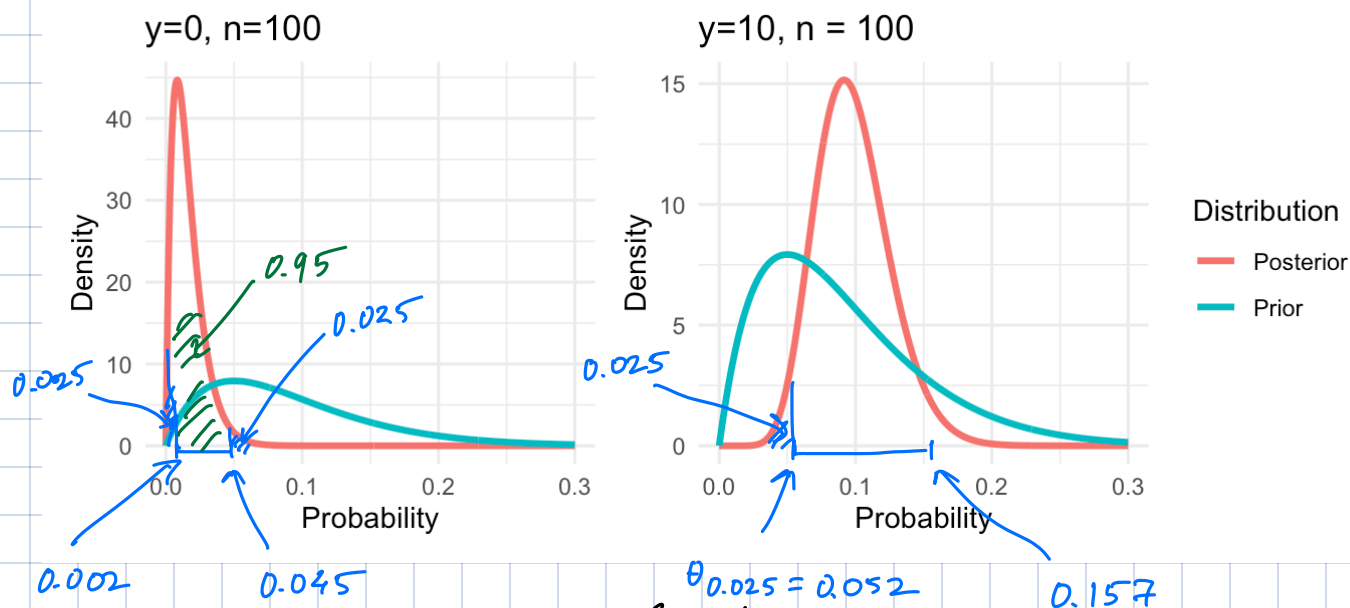
↑
posterior mean

More concretely, in our proportion estimation example

$$\theta | y \sim \text{Beta}(L+y, n-y+\beta) \Rightarrow E(\theta | y) = \frac{L+y}{L+y+n-y+\beta} = \frac{L+y}{L+\beta+n}$$

prior contribution

$$E(\theta | y) = \frac{2+0}{2+2+100} = \frac{2}{122} \approx 0.016 \quad E(\theta | y) = \frac{2+10}{2+2+100} = \frac{12}{122} \approx 0.098$$



2) uncertainty quantification:

Bayesian credible intervals for univariate parameters: $(\theta_{i, q/2}, \theta_{i, 1-q/2})$, where

$$\Pr(\theta_i < \theta_{i, q/2} | y) = q/2 \quad \text{and} \quad \Pr(\theta_i < \theta_{i, 1-q/2}) = 1 - \frac{q}{2}$$

$$\Pr(\theta_i > \theta_{i, 1-q/2}) = \frac{q}{2}$$

In our proportion estimation example:

$$y=0, n=100, L=2, \beta=20, q=0.05$$

this gives us 95% credible interval

$$(\theta_{0.025}, \theta_{0.975}) = (0.002, 0.045)$$

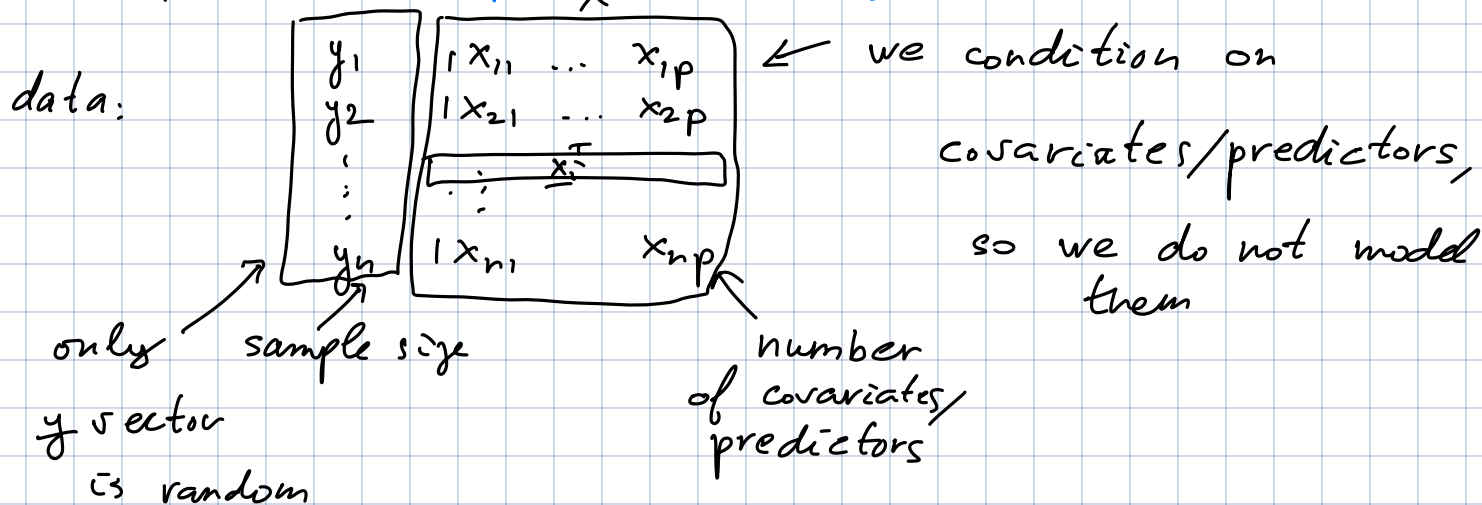
$$y=10, n=100, L=2, \beta=2, q=0.05$$

$$(\theta_{0.025}, \theta_{0.975}) = (0.052, 0.157)$$

There is no formula for quantiles, so these calcu-

lations must be done numerically (gheta?) in R)

Example: multiple linear regression



Data generating model: $y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i$,
where $\varepsilon_1, \dots, \varepsilon_n \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$ or
in vector form $y_i = \underline{x}_i^T \underline{\beta} + \varepsilon_i$

parameters: $\underline{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \sigma^2$

Likelihood: $p(\underline{y} | \underline{\beta}, \sigma^2) = \{\text{independence assumption}\} =$
 $= \prod_{i=1}^n p(y_i | \underline{\beta}, \sigma^2)$, where
 $p(y_i | \underline{\beta}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - \underline{x}_i^T \underline{\beta})^2}{2\sigma^2}}$

priors: many choices, but let's use something simple:
 $\beta_0, \beta_1, \dots, \beta_p \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$ constant of our choice
 $\sigma^2 \sim \text{Gamma}(\cdot, \cdot)$

posterior distribution / density:

$$p(\underline{\beta}, \sigma^2 | \underline{y}) = \frac{p(\underline{y} | \underline{\beta}, \sigma^2) p(\underline{\beta}) p(\sigma^2)}{\int \dots d\underline{\beta} d\sigma^2}$$

Bad news: Bayesian inference requires calculating high dimensional integrals

Good news: We have algorithms for this task

Monte Carlo Integration

First, let's review the concept of mathematical expectation:

1) for discrete random variable X

$$E[g(X)] = \sum_{k=1}^n g(x_k) P(X=x_k)$$

1st moment: $g(x)=x$

2nd moment $g(x)=x^2$

2) for (absolutely) continuous random variable X :

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Example: exponential random variable

$f(x) = \lambda e^{-\lambda x} \cdot \mathbb{1}_{\{x \geq 0\}}$, where $\lambda > 0$ - rate parameter

$$X \sim \text{Exp}(\lambda) \quad E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \cdot \lambda \cdot e^{-\lambda x} = \text{[integration by parts]} = \dots = \frac{1}{\lambda}$$

Strong Law of Large Numbers (SLLN)

Let $X_1, X_2, \dots$ be independent and identically distributed (iid) random variables with

$\mu = E(X_i) < \infty$ Then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = \mu$$

SLLN says that empirical averages of iid random variables converge to the theoretical average/expectation

Monte Carlo Integration

Objective: $E[h(X)] = \int h(x) f(x) dx$, where X is a random variable with probability density function $f(x)$ or

$E(h(X)) = \sum_{k=1}^{\infty} h(X_k) p_k$, where X is a discrete random variable with prob. mass function $p_1, p_2, \dots$

If $X_1, \dots, X_n \stackrel{iid}{\sim} f(x)$ and $E(h(X_i)) < \infty$, then

$$\text{SLLN} \Rightarrow \frac{1}{n} \sum_{i=1}^n h(X_i) \approx E(h(X_i))$$

$\overline{h_n}$

Example: second moment of the Beta distribution

Recall that in our estimating proportions example, one setting had:

$$\theta \sim \text{Beta}(2, 20)$$

$$y = 10, \quad n = 100$$

$$\theta | y \sim \text{Beta}(12, 110)$$

Objective : $E(\theta^2 | y)$ - second moment

In this case we can derive a formula :

$$X \sim \text{Beta}(\alpha, \beta) \Rightarrow E(X^2) = \frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)}$$

This result tells us that

$$E(\theta^2 | y) = \frac{12(12+1)}{(12+110+1)(12+110)} = 0.01$$

Let's see if we can get to the same number using Monte Carlo integration.

Our game plan :

- 1) generate $\theta_1, \dots, \theta_m \sim \text{Beta}(12, 110)$
- 2) return $\frac{1}{m} \sum_{i=1}^m \theta_i^2 \approx E(\theta^2 | y)$

computer demo [here](#)